Standing Waves on Strings: Harmonics, Tension and Densi

Standing Waves on Strings · oPhysics

Senior Secondary (Years 11-12)	~30 minutes	Science · Physics

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Simulation: https://ophysics.com/w8.html
Curriculum links: waves

Learning intentions

- I can relate the harmonic number of a standing wave to its frequency.
- I can use $v = f \cdot \lambda$ and $v = \sqrt{(T/\mu)}$ to predict a string's wave speed.
- I can explain how tension and linear density affect the frequency of a given harmonic.

Getting started

Open the simulation and set a fixed tension and linear density for the string. Adjust the frequency slider slowly until you see the first (fundamental) standing wave pattern form clearly.

Tasks

1. Find the frequency of the fundamental (1st harmonic) for your string, then find the frequencies of the 2nd, 3rd and 4th harmonics.

Harmonic number	Frequency

2. Using your table from Task 1, describe the pattern between harmonic number and frequency.

3. Keeping the frequency fixed at the fundamental you found, double the tension. Predict what will happen to the standing wave pattern, then check.

Working:

4. Using $v = \sqrt{(T/\mu)}$ and $v = f \cdot \lambda$, calculate the expected fundamental frequency for your string's tension and linear density, and compare it to the value you measured in Task 1.

Working:

Extension (optional)

Explain, using this relationship, why tightening a guitar string raises its pitch.